Task 1 Report: Training nanoGPT on ROCStories

COMP4680/8650 Mini-Project 1

This report summarises a scratch-trained nanoGPT baseline for ROCStories. The final reported run keeps the official baby-GPT architecture (6 layers, 6 heads, 384 hidden size; 29.94M parameters) and reaches average loss 3.216 and perplexity 24.93 on the public ROCStories test split used locally for model selection.

Pipeline correctness

ROCStories is processed with GPT-2 BPE via tiktoken. Each story receives an end-of-text token (50256), then the token streams are written to train.bin and val.bin. The public test split (19,633 stories) is used locally as the validation/evaluation split, and it is also exported as blank-line-separated test_full.txt so the exact result can be measured with the unmodified eval.py workflow.

Dataset statistics motivated a shorter context length: mean story length is 52.35 tokens, the 95th percentile is 69, and the maximum observed validation length is 91 tokens. This made block_size=96 a better fit than the earlier 128-token baseline.

Experimental rigor

Aspect	Value
Model	Scratch only; n_layer=6, n_head=6, n_embd=384, bias=False
Final config	batch=80, block=96, dropout=0.14, lr=3.5e-4, cosine decay, warmup=500, max_iters=12000, seed=2027
Compute	Single NVIDIA GeForce RTX 4060 Laptop GPU; compile=False for Windows stability

Results and qualitative behaviour

The best validation checkpoint in the final run was reached at step 11,175 with sampled validation loss 3.2774. Exact evaluation on the public ROCStories test split used 19,633 paragraphs and 988,345 predicted tokens, giving average loss 3.216 and perplexity 24.93.

Qualitatively, conservative decoding (temperature 0.7, top-k 40) produces more stable five-sentence arcs, while temperature 0.9 and top-k 100 increase diversity but also grammar slips and logic jumps. A representative low-temperature output for 'Emily forgot her umbrella before leaving for work' kept a coherent short-story structure, although it still repeated the rain motif.

Main strengths are improved ROCStories cadence and better low-temperature coherence relative to the original baseline. Typical failure modes remain repetition, awkward endings, and occasional drift into a second story if generation is not truncated at the end-of-text token.

Constraint check

- Scratch training only
- Official baby-GPT scale
- 29.94M parameters (<= 32M)
- Unmodified eval.py-compatible workflow

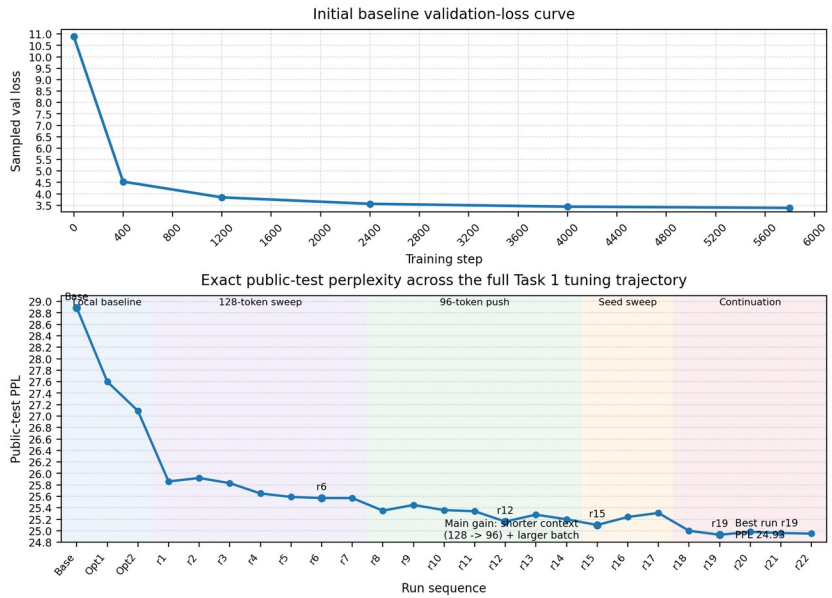

Figure 1. Top: sampled validation-loss curve for the original 6k-step baseline. Bottom: exact public-test perplexity across all reported Task 1 tuning runs, with more detailed axes and phase labels to show the step-by-step tuning process.

Metric	Value
Exact public test	avg_loss 3.216; PPL 24.93
Best tuning gain	PPL 28.89 -> 24.93